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Knowledge Check: What Is Deep Learning?
Practice Assignment • 10 min

Due Oct 23, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following is a potential drawback of using deep learning models on small datasets? 1 / 1 point

- They fail to classify structured data
- They require complex feature engineering
- They may overfit the training data.
- They produce very low accuracy on all types of tasks

Nice work

Correct. Deep learning models can overfit when trained on small datasets.

2. What is a common disadvantage of deep learning models? 1 / 1 point

- They perform poorly on large datasets
- They require feature engineering
- They are limited to processing structured data
- They lack interpretability.

Nice work

Correct. Deep learning models currently lack transparency in showing the importance of individual input features.

3. What is a key characteristic of traditional machine learning models compared to deep learning models? 1 / 1 point

- They rely heavily on feature engineering.
- They automatically discover the best representation of features.
- They require large amounts of training data to perform well.
- They integrate feature extraction and classification into a single process.

Nice work

Correct. Traditional machine learning models depend on manually engineered features to represent the data for the model.

4. In which domain have deep learning models become state of the art? 1 / 1 point

- Arithmetic computation
- Natural language processing
- File compression algorithms
- Traditional database indexing

Nice work

Correct. Deep learning models have advanced many tasks in natural language processing, such as parsing and semantic labeling.

5. Which of the following are advantages of deep learning models? 1 / 1 point

- [ ] They require very little data to perform well
- [ ] They are highly interpretable models
- [x] They can automatically learn feature representations.

Nice work

Correct. One major advantage of deep learning is its ability to learn feature representations automatically.

- [x] They integrate feature extraction and classification.

Nice work

Correct. Deep learning models jointly perform feature extraction and classification.